data_exploration

```
cars <- read.csv("../data/cars-1945-2020.csv")
head(cars)
```

```
##   id_trim Make Modle   Generation Year_from Year_to    Series   Trim Body_type
## 1       1   AC   ACE 1 generation      1993    2000 Cabriolet 3.5 MT Cabriolet
## 2       2   AC   ACE 1 generation      1993    2000 Cabriolet 4.6 MT Cabriolet
## 3       3   AC   ACE 1 generation      1993    2000 Cabriolet 4.9 AT Cabriolet
## 4       4   AC   ACE 1 generation      1993    2000  Roadster 2.9 AT
## 5       5   AC   ACE 1 generation      1993    2000  Roadster 2.9 MT
## 6       6   AC   ACE 1 generation      1993    2000  Roadster 3.0 MT
##   load_height_mm number_of_seats length_mm width_mm height_mm wheelbase_mm
## 1                              2      4420     1870      1300         2470
## 2                              2      4420     1870      1300         2470
## 3                              2      4420     1870      1300         2472
## 4
## 5
## 6
##   front_track_mm rear_track_mm curb_weight_kg wheel_size_r14
## 1           1590          1570           1510
## 2           1590          1570           1510
## 3           1590          1570           1440
## 4
## 5
## 6
##   ground_clearance_mm trailer_load_with_brakes_kg payload_kg
## 1                 130                                    NA
## 2                                                        NA
## 3                                                        NA
## 4                                                        NA
## 5                                                        NA
## 6                                                        NA
##   back_track_width_mm front_track_width_mm clearance_mm full_weight_kg
## 1                  NA                   NA                         NA
## 2                  NA                   NA                         NA
## 3                  NA                   NA                         NA
## 4                  NA                   NA                         NA
## 5                  NA                   NA                         NA
## 6                  NA                   NA                         NA
##   front_rear_axle_load_kg max_trunk_capacity_l
## 1                                         300
## 2                                         300
## 3                                         300
## 4
## 5
## 6
##   cargo_compartment_length_width_height_mm cargo_volume_m3
```

```
## 1
## 2
## 3
## 4
## 5
## 6
##   minimum_trunk_capacity_l maximum_torque_n_m             injection_type
## 1                      185               407 Multi-point fuel injection
## 2                      185               427 Multi-point fuel injection
## 3                      185               434 Multi-point fuel injection
## 4                       NA
## 5                       NA
## 6                       NA
##   overhead_camshaft cylinder_layout number_of_cylinders compression_ratio
## 1                          V-type                   8
## 2                          V-type                   8
## 3                          V-type                   8
## 4                                                  NA
## 5                                                  NA
## 6                                                  NA
##   engine_type valves_per_cylinder boost_type cylinder_bore_mm stroke_cycle_mm
## 1    Gasoline                   4      Turbo               83              81
## 2    Gasoline                   4                          90              90
## 3    Gasoline                   2                         101              76
## 4                              NA                          NA              NA
## 5                              NA                          NA              NA
## 6                              NA                          NA              NA
##   engine_placement cylinder_bore_and_stroke_cycle_mm
## 1
## 2
## 3
## 4
## 5
## 6
##   turnover_of_maximum_torque_rpm max_power_kw presence_of_intercooler
## 1                           4000           NA
## 2                           4600           NA
## 3                           3250           NA
## 4                                          NA
## 5                                          NA
## 6                                          NA
##   capacity_cm3 engine_hp engine_hp_rpm     drive_wheels bore_stroke_ratio
## 1         3505       354          6500 Rear wheel drive
## 2         4601       326          5800 Rear wheel drive
## 3         4942       260          5250 Rear wheel drive
## 4                     NA
## 5                     NA
## 6                     NA
##   number_of_gears turning_circle_m transmission
## 1               6               NA       Manual
## 2               5               NA       Manual
## 3               4               NA    Automatic
## 4              NA               NA
## 5              NA               NA
```

```
## 6              NA               NA
##   mixed_fuel_consumption_per_100_km_l  range_km emission_standards
## 1                                  NA 450|1,000
## 2                                  NA   600|900
## 3                                   8
## 4                                  NA
## 5                                  NA
## 6                                  NA
##   fuel_tank_capacity_l acceleration_0_100_km.h_s max_speed_km_per_h
## 1                   90                         5               250
## 2                   90                         5               250
## 3                   80                        NA               225
## 4                                             NA
## 5                                             NA
## 6                                             NA
##   city_fuel_per_100km_l CO2_emissions_g.km fuel_grade highway_fuel_per_100km_l
## 1                    20                NA         95                         9
## 2                    15                NA         98                        10
## 3                    NA                NA         95                        NA
## 4                    NA                NA                                   NA
## 5                    NA                NA                                   NA
## 6                    NA                NA                                   NA
##   back_suspension     rear_brakes    front_brakes front_suspension
## 1 Helical springs ventilated disc ventilated disc  Helical springs
## 2 Helical springs ventilated disc ventilated disc  Helical springs
## 3 Helical springs ventilated disc ventilated disc  Helical springs
## 4
## 5
## 6
##   steering_type car_class country_of_origin number_of_doors safety_assessment
## 1                                                        NA
## 2                                                        NA
## 3                                                        NA
## 4                                                        NA
## 5                                                        NA
## 6                                                        NA
##   rating_name battery_capacity_KW_per_h electric_range_km charging_time_h
## 1                                                      NA
## 2                                                      NA
## 3                                                      NA
## 4                                                      NA
## 5                                                      NA
## 6                                                      NA
```